Algebra 1 Name____________________
Unit 11 Date____________________
Lesson 3 Homework Period__________________

For questions 1 – 2, consider the function $f(x) = -4(1 + 0.65)^x$.

1. Since a is
- o positive
- o negative

and b is
- o between 0 and 1,
- o greater than 1,

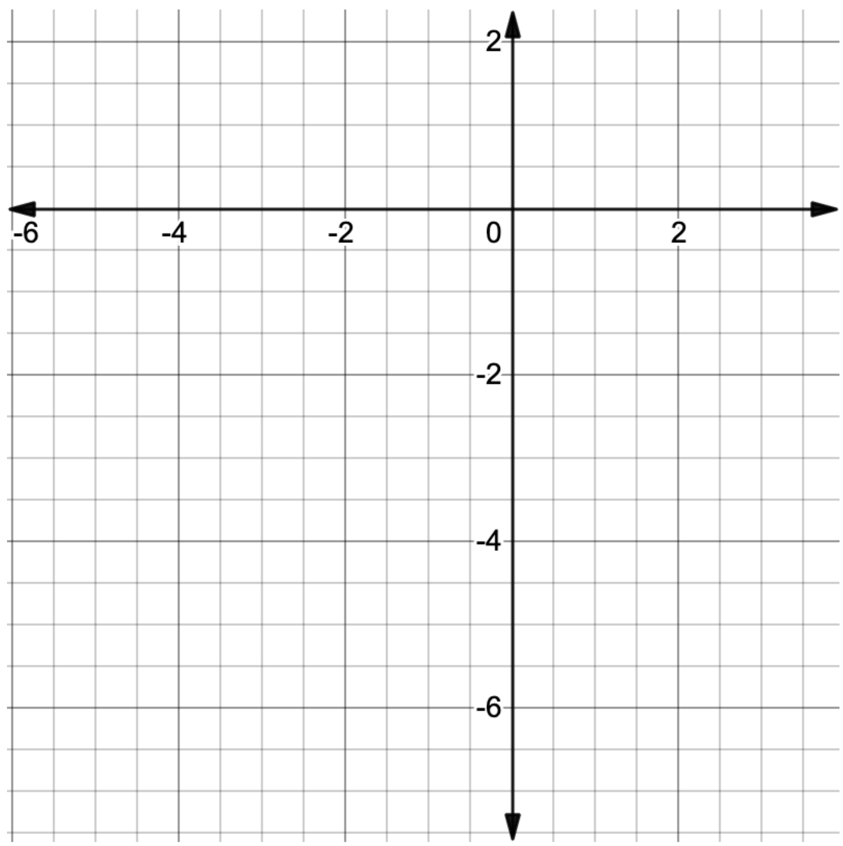
the function is
- o increasing.
- o decreasing.

The left end behavior of the graph is
- o $y \to 0$
- o $y \to -\infty$

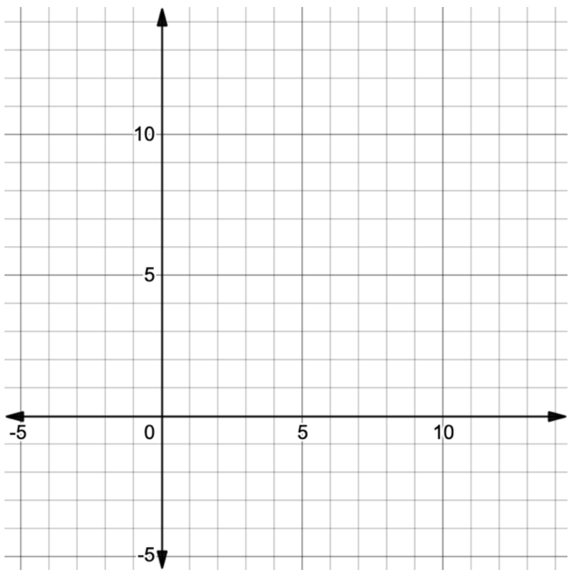
and the right end behavior is
- o $y \to 0$.
- o $y \to -\infty$.

The y-intercept of the function is
- o -4.
- o 0.65.
- o 1.65.

2. Sketch the graph of $f(x) = -4(1 + 0.65)^x$.



Graph the function of $y = 5\left(\frac{2}{3}\right)^x$, complete the table of values, and then determine all key features.

3. Graph	4. Table of Values
	see table below

x	y
-2	
-1	
0	
1	
2	

Key Features	
5. Growth or Decay	
6. Domain and Range	
7. Increasing/Decreasing	
8. Constant Percent Rate of Change	
9. Left and Right End Behavior	
10. Positive or Negative	